



CSC1029 OO Programming

Lecture 28 – Further Recursion Examples
[... and a look ahead to Assessment 3]



What's to come in today's lecture?

- Recursion Examples
- Preparing for Assessment 3

Recursive Functions (Methods)

```
public class Calculator {  
  
    // x multiplied by y for x,y >=0  
    public static int multiply(int x, int y) {  
        return -1;  
    }  
  
    // x divided by y for x>=0 and y>0  
    public static int divide(int x, int y) {  
        return -1;  
    }  
  
    public static void main(String[] args) {  
        int X = 4;  
        int Y = 3;  
  
        System.out.println(X + " multiplied by " + Y + " = " + multiply(X,Y));  
        System.out.println(X + " divided by " + Y + " = " + divide(X,Y));  
    }  
}
```

Recursive Functions (Methods)

```
// x multiplied by y for y >=0
public static int multiply(int x, int y) {
    // begin by looking at the problem
    // multiplication may be defined by a sequence
    // of repeated additions:
    // multiply X by Y
    // ->> equivalent to X added to itself Y times?
    // ->> multiply 4 by 3
    // ->> 4 + 4 + 4
    // ->> 4 + multiply 4 by 2 etc
}
```

Recursive Functions (Methods)

```
// x multiplied by y for y >=0
public static int multiply(int x, int y) {

    // So - we need to define a BASE CASE
    if ( y == 0 ) {
        return 0;
    }

    // And - we need to define a RECURSIVE STEP
    return x + multiply(x, y-1);
}
```

Recursive Functions (Methods)

```
// x divided by y for x>=0 and y>0
public static int divide(int x, int y) {
    // may be defined by a sequence of repeated subtractions
    // divide X by Y (i.e. how many times does Y occur in X - integer division
    // i.e. divide 6 by 2 → Same as 1 + divide 4 by 2
    // BASE CASE
    if ( y <=0 || x < 0 ) {
        return -1 ; // error, can't divide by 0 or less
    }
    if ( x < y ) {
        return 0;
    }

    // RECURSIVE STEP
    return 1 + divide(x-y, y);
}
```

Examination Details

When :	Wednesday 26th April 2023
	During scheduled lab : 3pm – 5pm
How:	Canvas Quiz
How Long:	1 hour 15 minutes
How many questions:	~30
Open Book:	Yes (restricted to Canvas material only, Eclipse will NOT be permitted)
Can be taken remotely:	No – MUST be taken in the lab

Examination Structure:

How are questions typically structured in 'Canvas'?

- **multiple choice** – select one correct answer from a list of options
- **multiple choice** – select the correct answer(s) from a list of option
- **free text entry** – enter text into a text box

Examination Structure: Possible Topics

1. **OOP Fundamentals**
[instance data & behaviour, class data & behaviour, instantiation, construction]
2. **Class Design & Implementation**
[UML- classes & associations, User input & exception handling]
3. **Object Collections**
[Arrays & ArrayLists]
4. **Polymorphism**
[Runtime vs Compile-time, Inheritance, Code reuse, Method overloading & overriding]
5. **Searching Algorithms**
[Searching sorted & unsorted data]
6. **Sorting Algorithms**
[Insertion sort, Bubble sort]
7. **Fundamental Data Structures**
[Linked lists, stacks, queues & associated algorithms]
8. **Recursion**
[Recursive algorithms]
9. **Testing**
[Equivalence Partitioning, Boundary Value Analysis]



What's Next?

A Revision Session – (Via Teams) Tuesday April 18th
(Send me emails on topics you would like me to cover)

Assessment 3 – (Canvas Quiz, in the usual lab session) Wednesday April 26th

